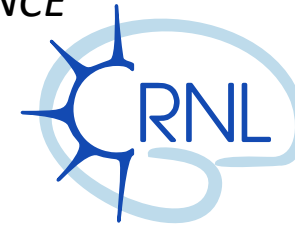


To what extent can cognitive and motor skills be dissociated in DoC patients?

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EMBC 2025 - Copenhagen

July 16th workshop – *Exploring Altered States of Consciousness Through EEG and Brain-Computer Interfaces*

Cognitive-Motor Dissociation (CMD)

“Cognitive Motor Dissociation (CMD) describes a condition whereby brain injury patients can demonstrate response to command through advanced electrophysiology and imaging assessments but are unable to do so through in standard, behavior-based, clinical assessments”

Monti, medRxiv 2025

Outline

- Introduction
- Detecting Cognitive Motor-Dissociation (CMD)
- From Detecting CMD to instantiating communication using BCI
- The limits of CMD

Introduction



- **Patients with disorders of consciousness**

Clinical states: coma / unresponsive wakefulness syndrome (UWS) / Minimally conscious state (MCS)

Etiologies: trauma, stroke, ...

- **Clinical challenge**

Functional evaluation: consciousness, preserved functions ?

Prognosis: how the patient will evolve (awakening, functional state) ?

Communication: if possible!

- **Assessments**

Clinical scales: CRS-R, Glasgow Outcome Scale, FOUR

Introduction



- **Patients with disorders of consciousness**

Clinical states: coma / unresponsive wakefulness syndrome (UWS) / Minimally conscious state (MCS)

Etiologies: trauma, stroke, ...

- **Clinical challenge**

Functional evaluation: consciousness, preserved functions ?

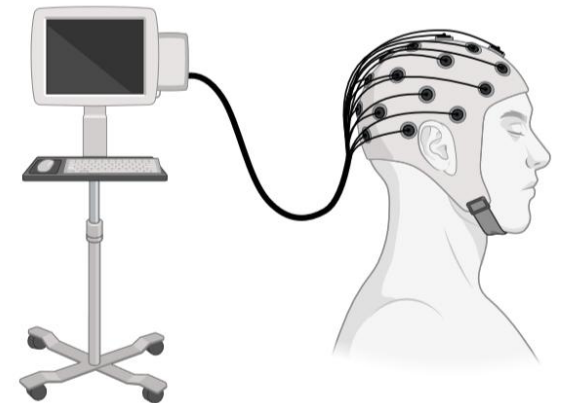
Prognosis: how the patient will evolve (awakening, functional state) ?

Communication: if possible!

- **Assessments**

Clinical scales: CRS-R, Glasgow Outcome scale, FOUR

Neurophysiological measures: bypass behavior, objective, quantitative (EEG, fMRI, PET)



Electroencephalography (EEG)

Introduction

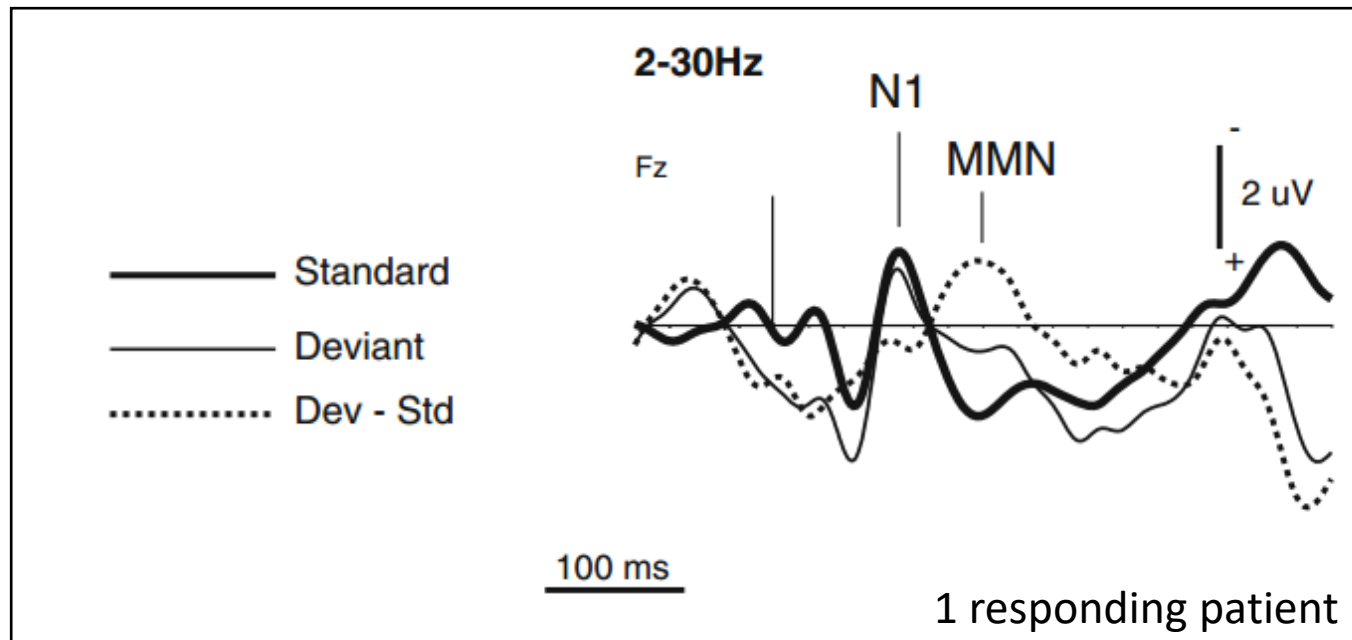
- Example of EEG markers of (automatic) information processing

Oddball paradigm

Deviant tone
Standard tone



Mismatch Negativity (MMN): Dev - Std



Introduction

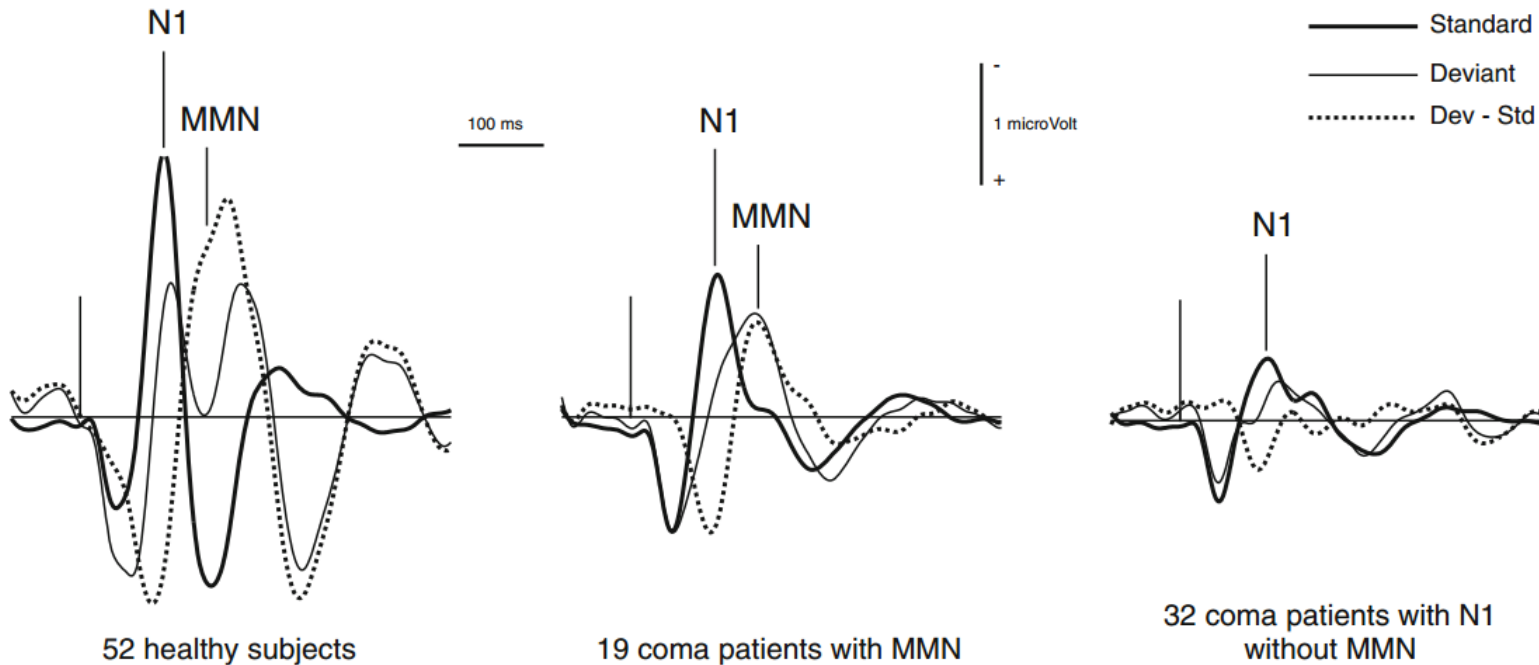
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Mismatch Negativity (MMN): Dev - Std



Introduction

- Example of EEG markers of (automatic) information processing

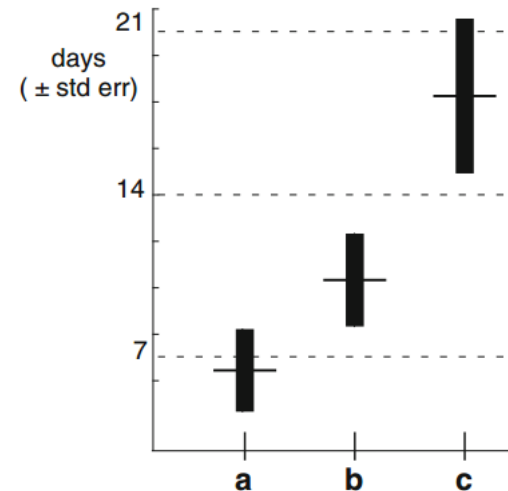
Oddball paradigm

Deviant tone
Standard tone



Mismatch Negativity (MMN): Dev – Std

Mean delay
between recording and awakening
(*n* = 95 patients who awakened)



- a: N1 and MMN present (*n* = 30)
- b: N1 present and MMN absent (*n* = 40)
- c: N1 and MMN absent (*n* = 25)

**MMN is a highly specific predictor
of coma awakening**

Introduction

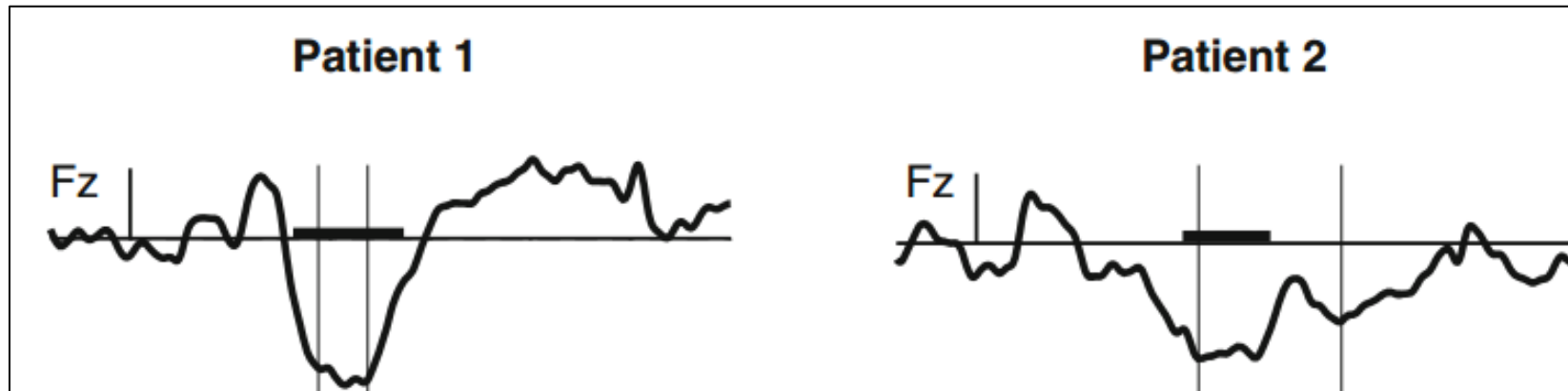
- **Example of EEG markers of (automatic) information processing** « Patient's own first name »

Novelty stimulus

Deviant tone
Standard tone



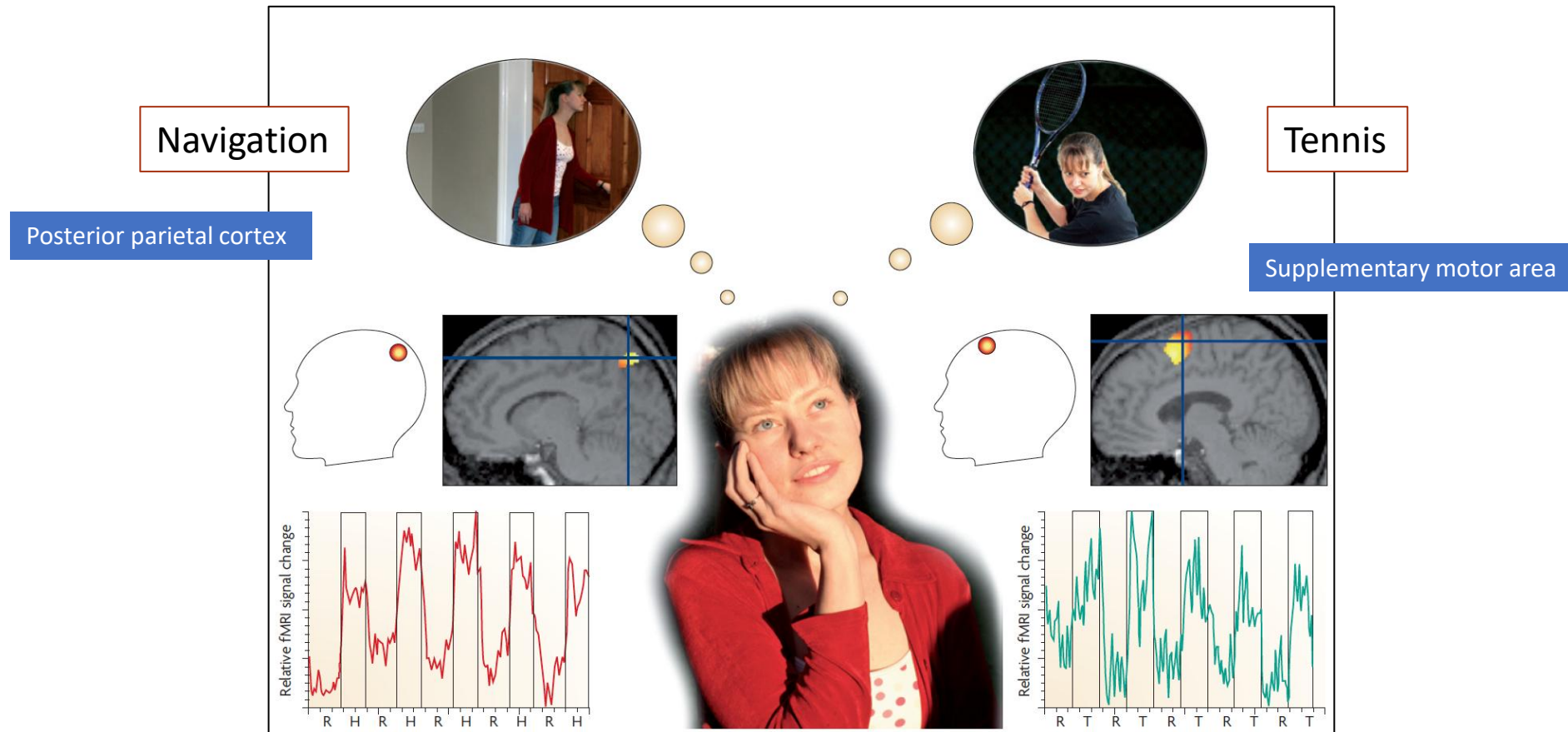
Attention orienting toward a rare, relevant stimulus: P3



Presence of a novelty P3 has shown a good correlation with coma awakening

Detecting Cognitive-Motor Dissociation

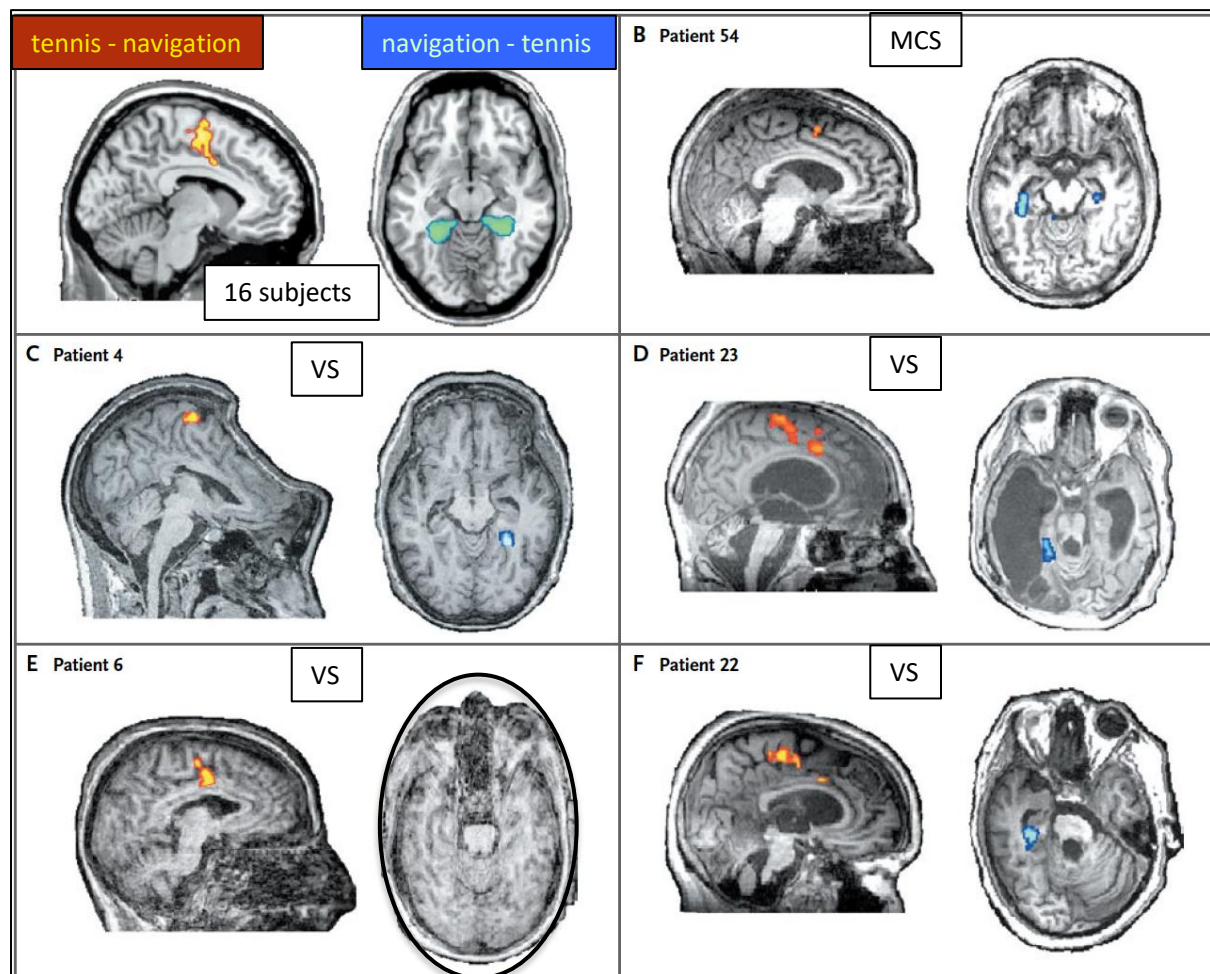
- From passive to active paradigms



Detecting Cognitive-Motor Dissociation

- From passive to active paradigms

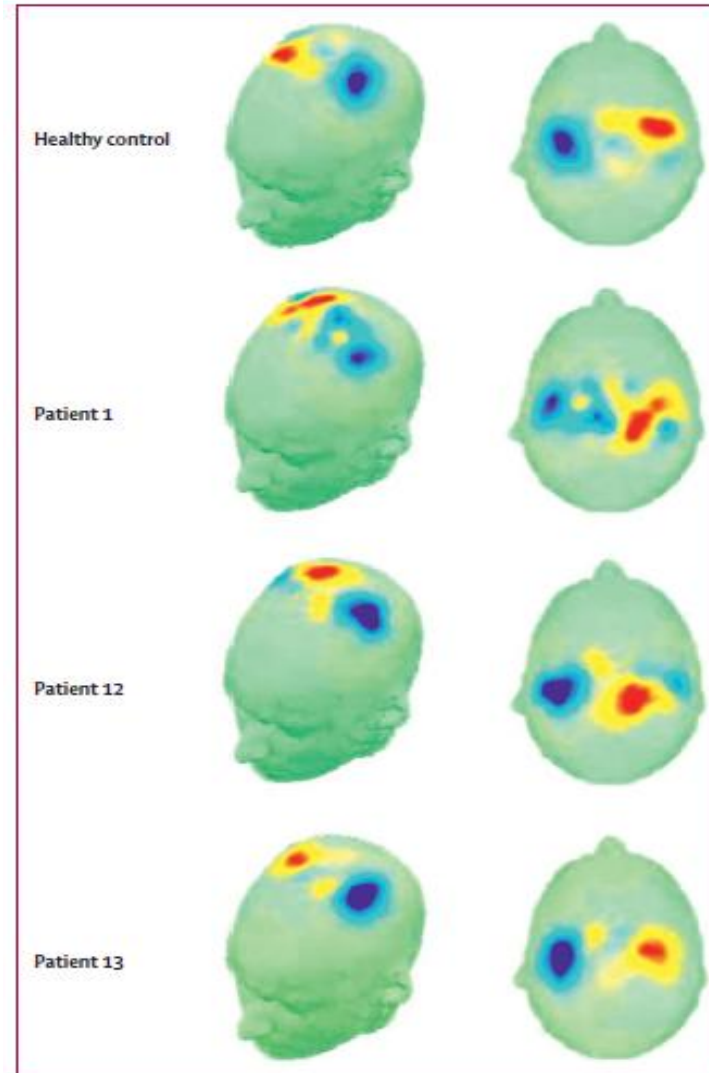
- 54 patients tested with fMRI
- 5 « responders »



Detecting Cognitive-Motor Dissociation

- EEG active paradigms in chronic patients

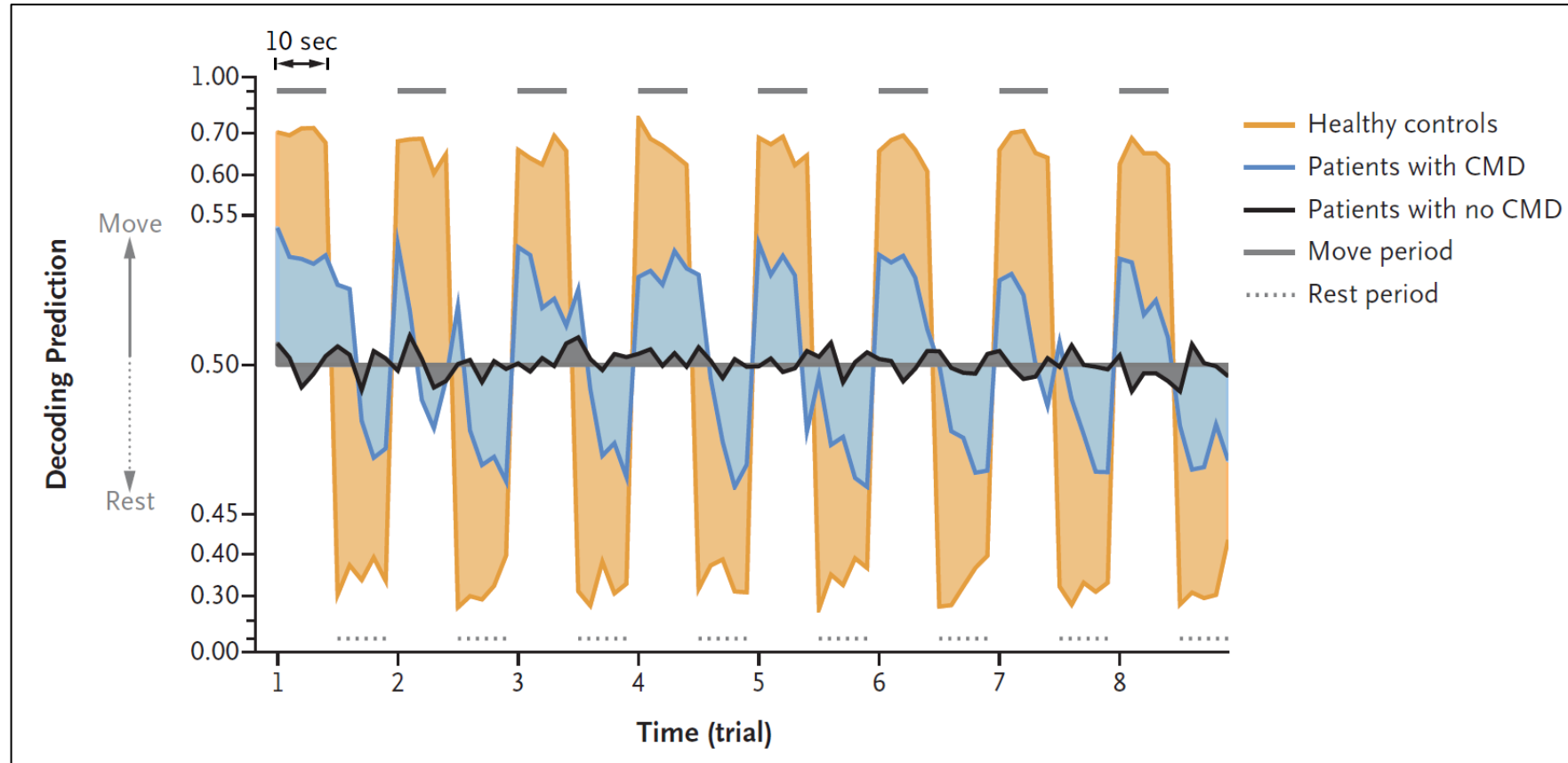
- 16 patients testés en EEG
- 3 « répondeurs »



Detecting Cognitive-Motor Dissociation

- EEG active paradigms in coma patients

- “Give me a handshake”
- 16 responders among 104 patients
- This response predicts a significant functional improvement

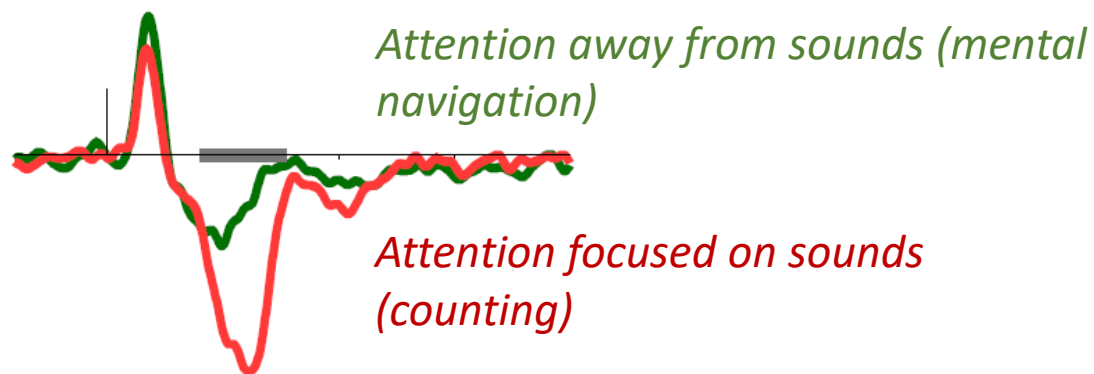


Detecting Cognitive-Motor Dissociation

- Active oddball paradigms



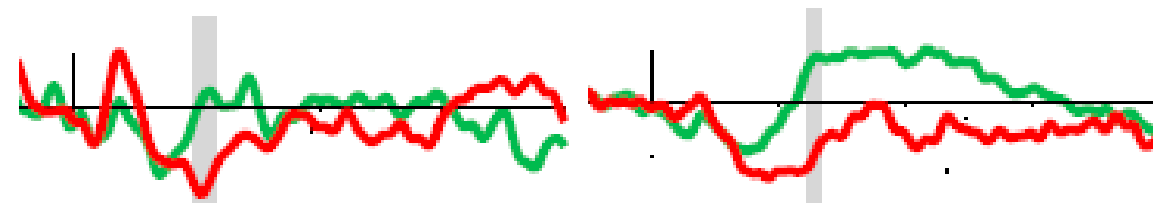
20 healthy controls



- EEG evoked responses are modulated by voluntary attention
- This effect can be observed at the individual level

Patient 3

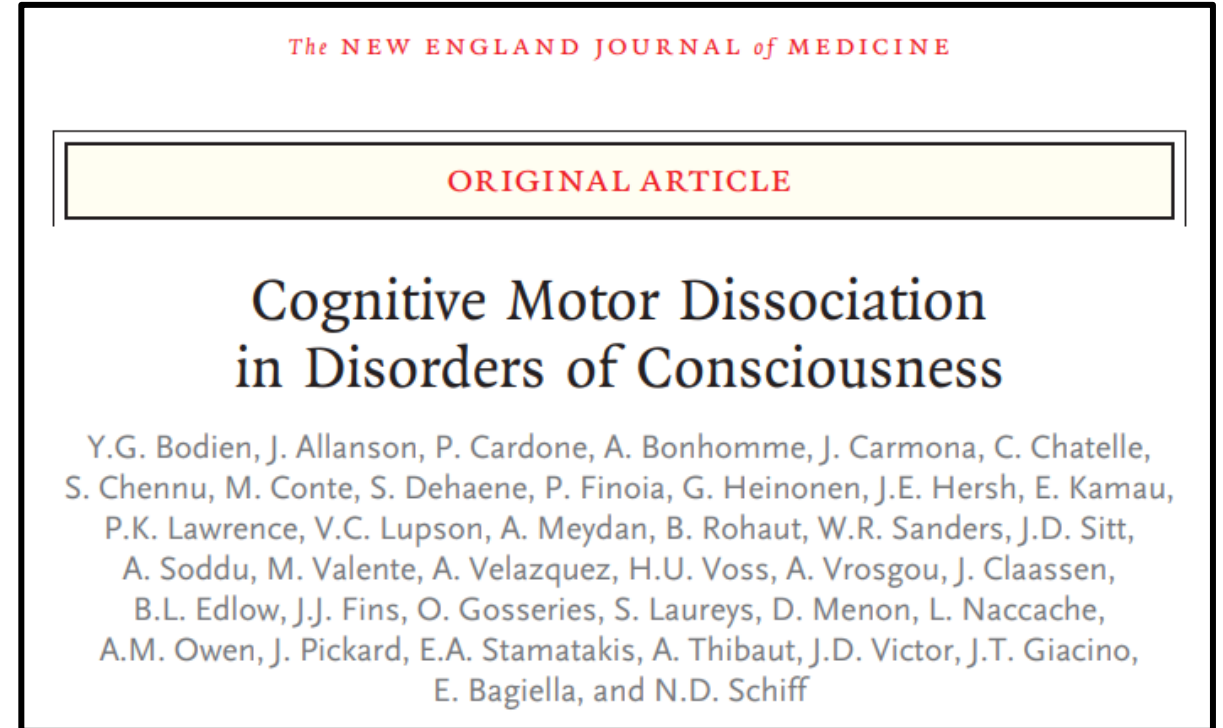
Patient 5



- 14 responders out of 63 patients (22%)
- 9/14 were in coma
- 93% of the responders were awake at 6 month, for only 51% of the non-responders

Detecting Cognitive-Motor Dissociation

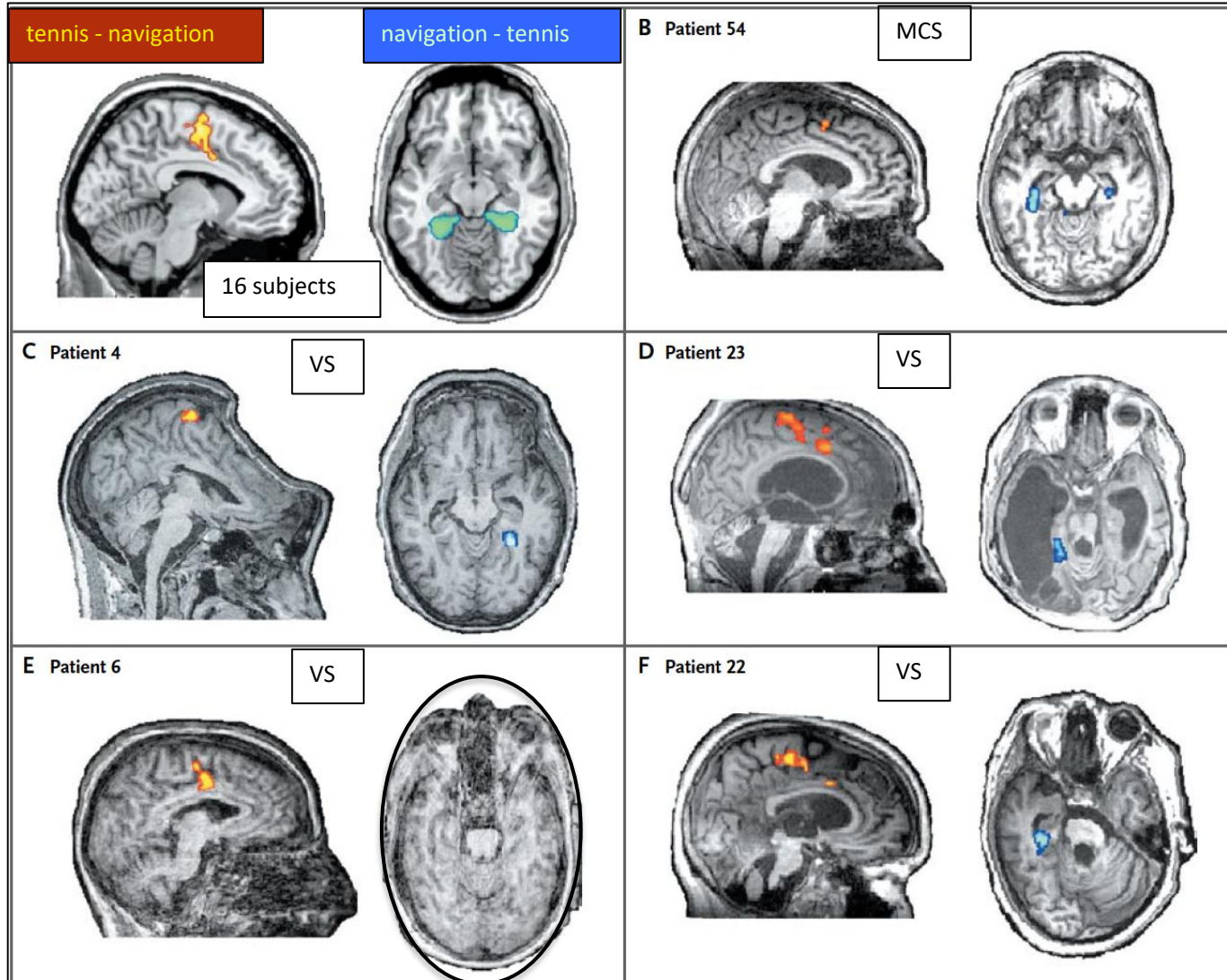
- Large cohort study with fMRI & EEG



- 241 patients with no observable behavioural response were tested with fMRI or/and EEG ; 60 (25%) showed a response with either fMRI or EEG
- 112 patients with an observable behavioural response were also tested, 38% responded

From detecting CMD to instantiating communication

From detecting CMD to instantiating communication



- 54 patients tested with fMRI
- 5 « responders »

Only 1 responding patient then managed to correctly answer closed questions

From detecting CMD to instantiating communication

- An auditory BCI for communication

Sound
« Non »

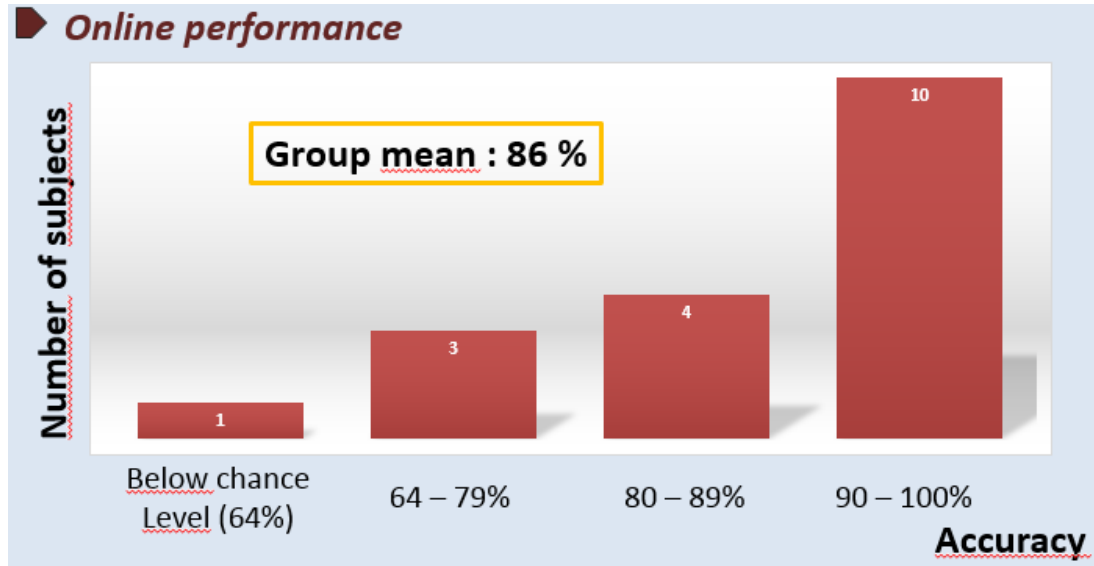


Sound
« Oui »



18 healthy volunteers

7 patients with a yes/no communication code



	Etiology	Age	Pathology duration	Offline Accuracy
LIS 1	Haemorrhagic stroke	60	3 y	53%
LIS 2	Traumatic	35	3 y	50%
LIS 3	Ischemia (post-trauma)	63	32 y	56%
ALS 1	Bulbar	53	6 y	91%
ALS 2	Limbs onset	31	5 y	51%
ALS 3	Limbs onset	55	4 y	100%
ALS 4	Limbs onset	60	25 y	100%

From detecting CMD to instantiating communication

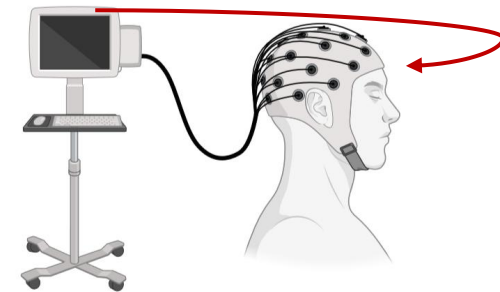
- Interim conclusion

Active paradigms and BCI protocols for communication are very similar

- EEG based
- Two class problem
- Both are challenging (no 100% control in conscious people)
- Both are inconclusive for non-responders (absence of proof is no proof of absence)
- Poor performances in patients, when used for communication

But BCI operates online

- It deals with the challenge of processing the data in real-time
- It provides feedback
- It has rarely been used in DoC patients



The limits of CMD

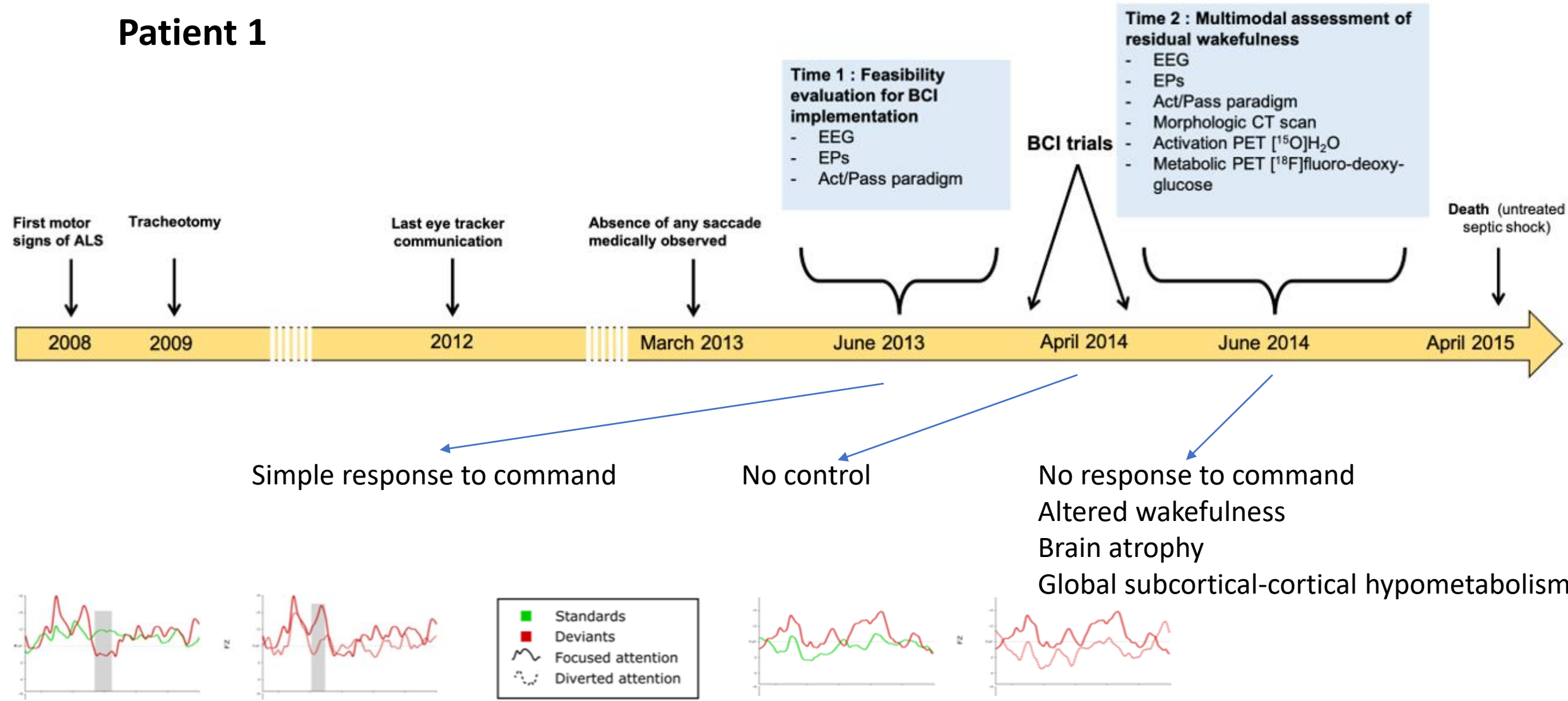
The limits of CMD

- BCI is considered the last means of communication for complete Locked-in state (cLIS) patients
- cLIS is rare, it may be observed at a late stage of ALS
- We shift the perspective from cLIS (presumed consciousness) to CMD (to-be-demonstrated consciousness)
- We performed a multimodal assessment of 2 ALS patients after they had lost their last means of communication (oculomotor function)

- ActPass auditory oddball
- Auditory BCI
- Wakefulness EEG
- Brain morphology
- Activation PET (Oxygen)
- Brain metabolism PET (Glucose)

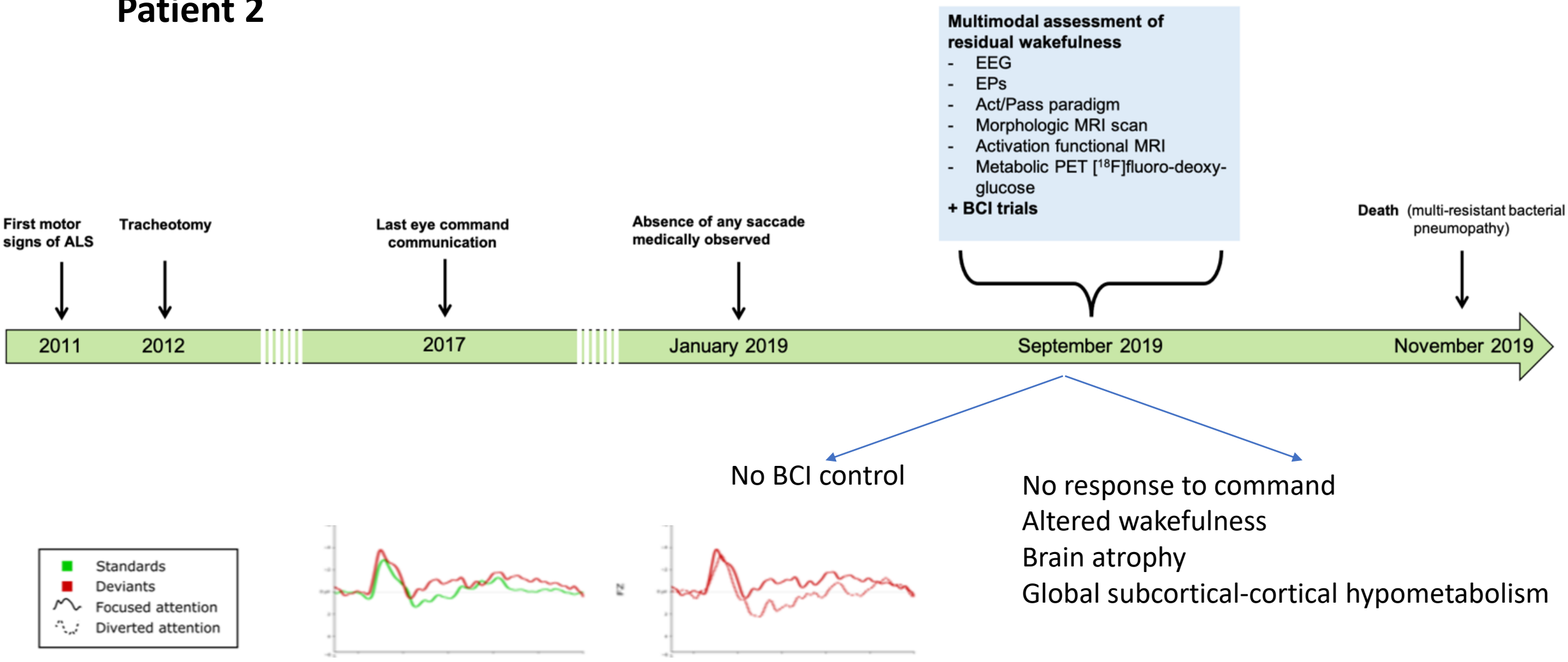
The limits of CMD

Patient 1



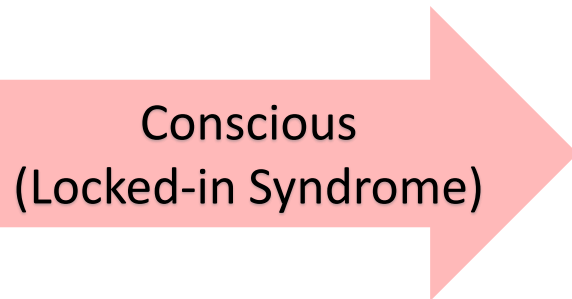
The limits of CMD

Patient 2



The limits of CMD

- These rare observations in a few patients are rather compatible with a DoC state, an extreme form of fronto-temporal dementia
- We cannot firmly demonstrate the absence of awareness, but each independent measure here concurs with suggesting that a « degenerative DoC » rather than cLIS might be the final stage of ALS
- Of course, this might not be the case for all ALS patients at a late stage, but this is a challenge for BCI !



BCI is ineffective for a substantial proportion of individuals with severe speech and physical impairment

Peters et al, 2022 ; Lesenfants et al, 2014 ; Seguin et al, 2024



No (longitudinal) reliable communication

Spüler, 2019 ; Galiotta 2022

The limits of CMD

- **Working hypothesis**

- BCI attempts to replace overt motor control by a covert attentional one
- There is evidence for a tight relationship between motor and attentional control in terms of functional anatomy in the brain, namely for spatial attention

Nobre, 2000; More & Fallah, 2004; Lowet, 2018

- > Severe (oculo-)motor impairment goes along with (covert) attention impairment
- > Extreme motor impairment (including oculomotor one) is associated with a lack of BCI control

Seguin, ..., Mattout, arxiv 2023 ; Under review

The limits of CMD

- Participants: 3 groups according to functional state

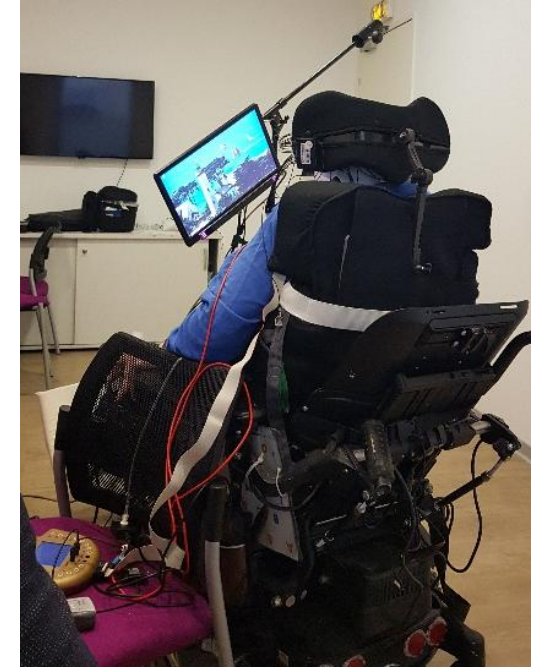
*Most severe
paralysis*

Locked-in
n = 19
(cannot talk, cannot write)
Can move their eyes
reliably enough to
communicate.

Tetraplegia
n = 28

Healthy subjects
n = 30

*Normal
motricity*



The limits of CMD

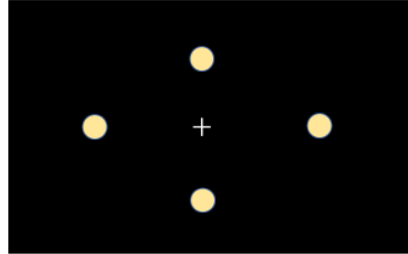
Seguin, ..., Mattout – Preregistrated in 2023

[OSF Registries](#) | [Assessment of voluntary control of attention in patients with severe motor impairment](#)

• Study protocol

Session 1

Target reaching



Task 1

Smooth pursuit



Task 2

Selective visual covert attention
(BCI-like)



Task 3

Auditory BCI



Task 4

30 min

30 min

Body spatial attention



Task 5

25 min

Deviant tone
Standard tone



Task 6

25 min

Session 2



+ Resting state

Task 7

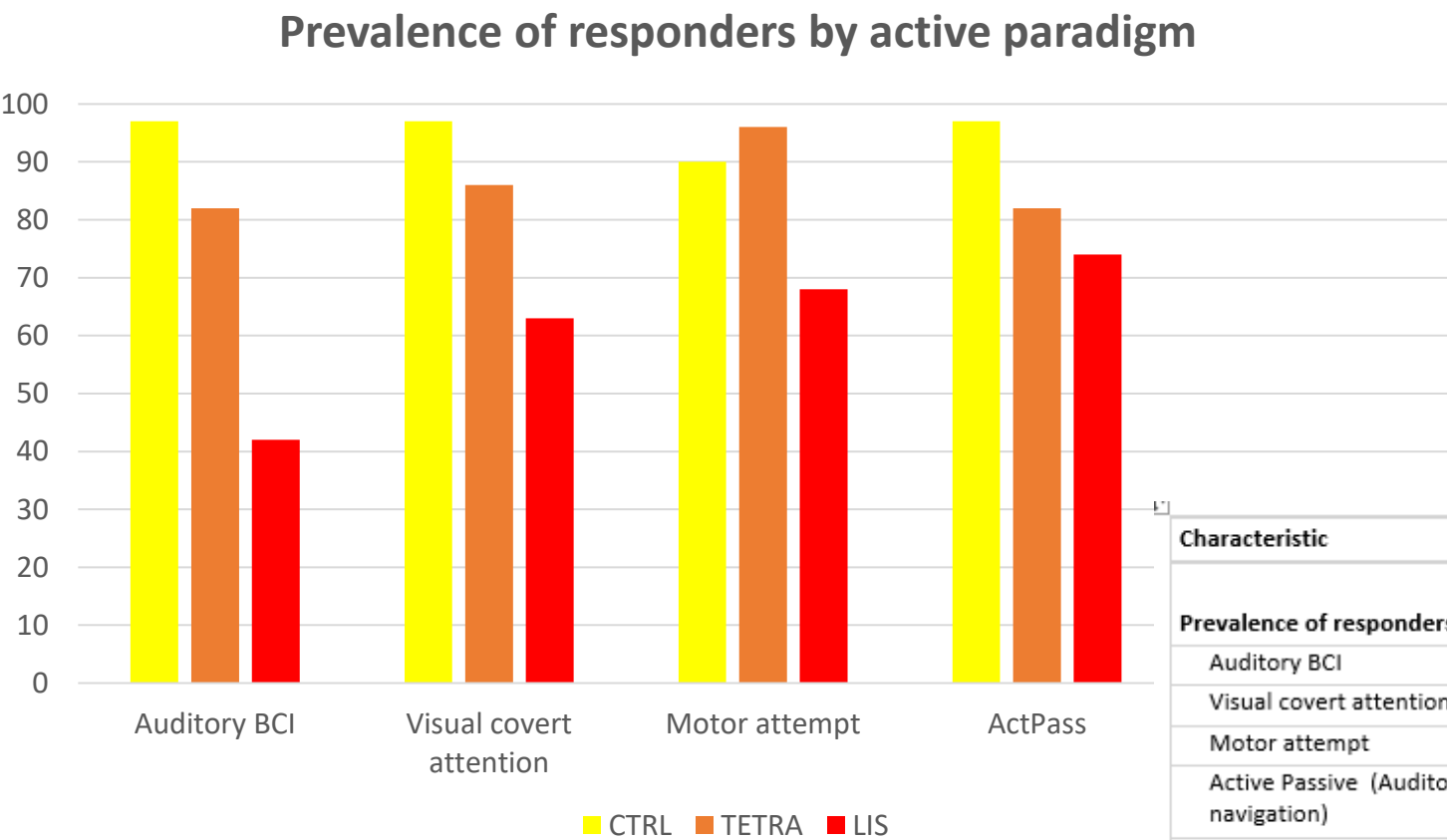
30 min

Clinical scales: motor abilities (ALS-FRS), neuropsychological abilities (BELIS)

Session 3

The limits of CMD

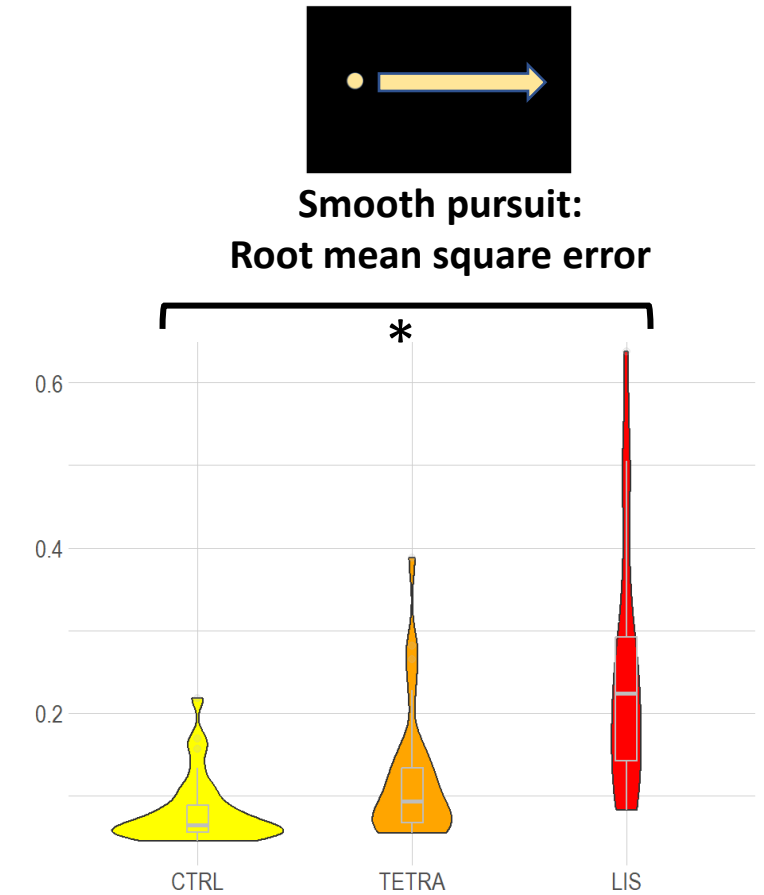
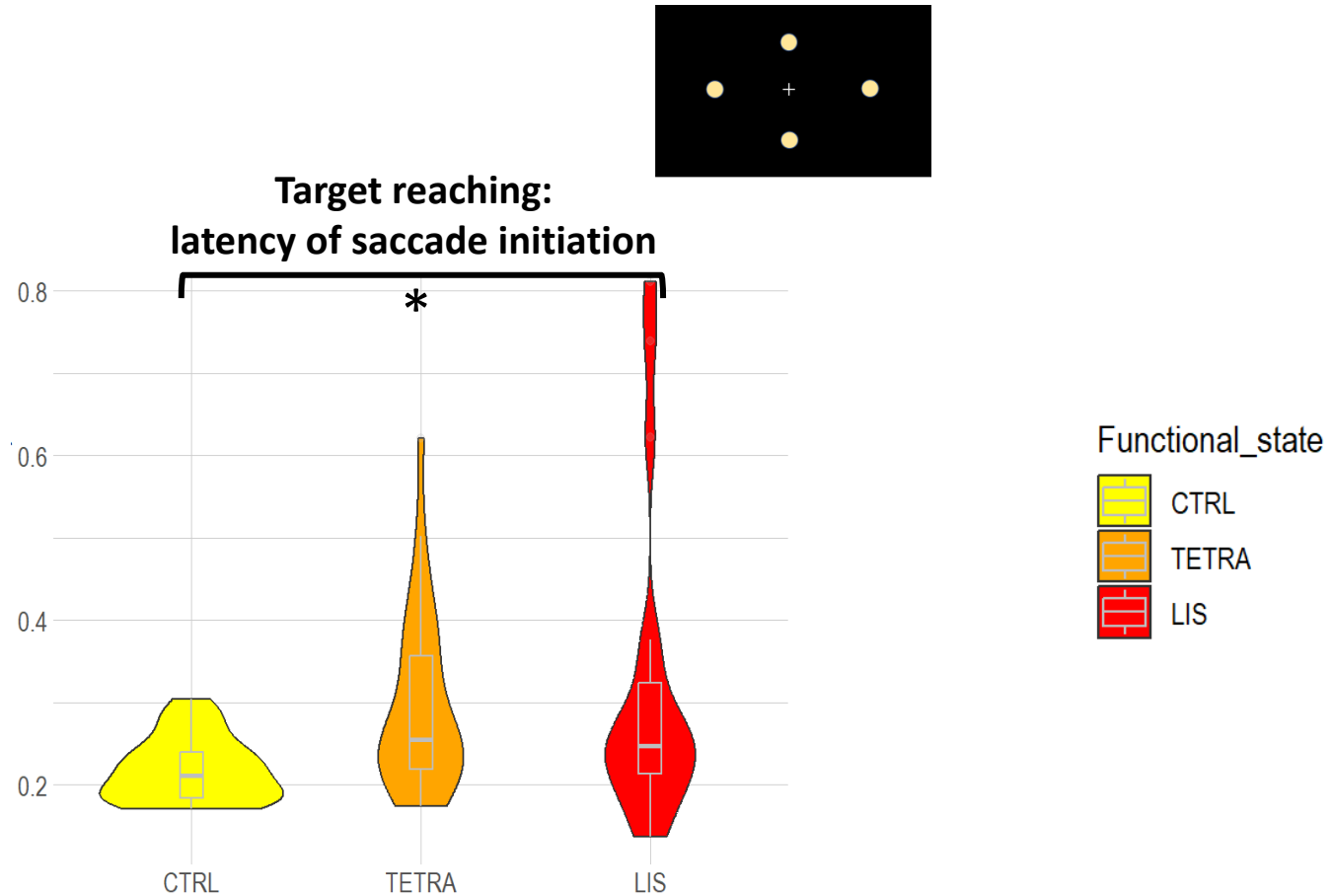
- Preliminary results: Group differences in EEG protocols



Characteristic	CTRL N = 30 ¹	TETRA N = 28 ¹	LIS N = 19 ¹	p-value ²
Prevalence of responders by active paradigm				
Auditory BCI	29 (97%)	23 (82%)	8 (42%)	<0.001**
Visual covert attention	29 (97%)	24 (86%)	12 (63%)	0.008*
Motor attempt	27 (90%)	27 (96%)	13 (68%)	0.024*
Active Passive (Auditory attention versus Mental navigation)	29 (97%)	23 (82%)	14 (74%)	0.044*
Number of paradigms with significant attentional modulation (median)	4 (4, 4)	3 (3, 4)	3.00 (1, 3)	<0.001**

The limits of CMD

- Preliminary results: Group difference in oculomotricity



The limits of CMD

- **Preliminary findings**
 - EEG: We found a significant difference between groups, mostly for the auditory BCI
 - Oculomotricity: We found a significant difference between groups, mostly for smooth pursuit

Functional state (clinical scales) + Smooth pursuit ability seem to best predict control performance of the auditory BCI control

Some conclusions & perspectives

- Problematics of DoC & LIS patients, of CMD detection & BCI communication converge
- Moving from CMD detection to communication is not trivial
- On average, BCI works poorly in patients who need it the most

Seguin, ... Mattout, arxiv 2023

Galiotta et al. Frontiers 2022

- What is CMD ?

Monti, medRxiv 2025

- What is required for BCI communication to be effective ?

Seguin, ... Mattout, arxiv 2023

Finally assessing patients' oculomotricity and attention abilities seems crucial to design and propose optimal BCI solutions

Acknowledgments

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Romain Bouet



Loriana Rizzo



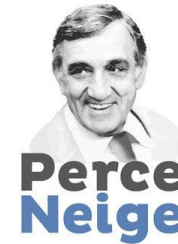
Lucie Gattaz



Ambre Query

**To all participants, their families and
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